



Kanformer: an attention-enhanced deep learning model for predicting student performance in virtual learning environments

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Abstract

With the rapid growth of digital education, massive open online courses (MOOCs) have gained widespread popularity. However, high failure rates and low completion rates remain challenges, highlighting the need for accurate student performance prediction to identify at-risk students and implement timely interventions. Many existing models assume simplistic or linear dependencies, failing to capture the complex relationships and cross-dimensional interactions between different student features. Therefore, KANFormer model is developed, a novel deep learning (DL) architecture that integrates multi-head self attention (MHSA) mechanisms with Kolmogorov–Arnold networks (KAN) to effectively captures complex patterns within large-scale student datasets, enabling more accurate predictions of student performance. KANFormer processes student demographic, academic, and engagement features from the open university learning analytics dataset (OULAD) while applying a feature elimination process to enhance dataset quality, reduce redundancy, and mitigate multicollinearity, ensuring optimal predictive performance. The model is evaluated on both binary and multi-class classification tasks, distinguishing successful students (“Distinction” & “Pass”) from unsuccessful ones (“Fail” & “Withdrawn”), while also predicting individual performance categories. Ablation experiments analyze the impact of MHSA and KAN on model performance. Experimental results on the OULA dataset, comprising over 29,000 samples, demonstrate that KANFormer outperforms traditional and state-of-the-art models, achieving superior accuracy, precision, and F1-score in student performance prediction. To enhance interpretability, SHAP (SHapley Additive exPlanations) values are utilized to quantify each feature’s contribution to predictions. SHAP analysis shows that academic assessments (exam scores, TMAs, CMAs) are the most influential predictors of student success, while engagement indicators play a secondary role. Furthermore, the model’s generalization ability is validated on additional datasets, including ASSISTments2012, xAPI, and the Mathematics and Portuguese Language Courses dataset, proving its scalability and robustness across different educational contexts. By accurately identifying at-risk students, KANFormer offers a scalable, interpretable, and highly effective solution for improving student retention and success in online education, providing actionable insights for personalized learning interventions.

Keywords Deep learning · Multi-head attention · Kolmogorov–Arnold networks · Student performance · Student outcome · Student success · Prediction

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1 Introduction

The rise of online education platforms, particularly Massive Open Online Courses (MOOCs), has significantly expanded access to learning. However, these platforms continue to struggle with high dropout rates and low completion rates (Annan and Asiedu 2024). A significant proportion of students fail or withdraw due to the lack of personalized support and timely intervention mechanisms, making it imperative to develop accurate student performance prediction models (Capuano et al. 2021; Yang and Bai 2022). Such models can enable educators and institutions to identify at-risk students

early and implement targeted interventions aimed at improving student retention and academic success. While research in educational data mining has explored various machine learning (ML) and deep learning (DL) approaches for student performance prediction, existing methodologies still present several critical limitations that reduce their practical applicability.

Traditional ML models, including Linear Regression, Logistic Regression, and Support Vector Machines, have been widely used to predict student success based on engagement metrics, assessments, and demographic features. However, these models typically assume linear relationships among features, making them inadequate for capturing the complex, multi-dimensional dependencies that characterize student learning behaviors (Alnasyan et al. 2024). For example, a student who frequently watches lecture videos while not actively participate in discussions may still succeed, whereas another student who frequently engages in quizzes while lacks conceptual understanding may struggle (Kuhlmann et al. 2024). These models fail to consider the interplay between passive engagement (e.g., watching videos) and active engagement (e.g., completing quizzes and assignments), which can vary significantly based on the subject matter and individual learning profiles (Chen and Zhai 2023). To address some of these limitations, tree-based models such as Decision Trees and Gradient Boosting Machines have been adopted. While these models handle non-linearity more effectively by capturing interactions between features, they still face challenges when modeling deep feature relationships and require extensive hyperparameter tuning to optimize their performance in high-dimensional datasets.

In recent years, DL models, particularly Transformer-based architectures, have demonstrated strong predictive capabilities in modeling sequential dependencies within time-series engagement data, such as clickstream logs and daily student interactions (Capuano et al. 2021; Zhang et al. 2023). However, the application of these models to structured student datasets, where features are aggregated rather than sequential, remains a challenge. Standard Transformer architectures are designed to capture temporal dependencies, making them less effective at learning complex cross-feature relationships within student performance data. Although some transformer-based models incorporate multi-layer perceptrons (MLP) and artificial neural networks (ANNs) into their architecture to enhance non-linearity, they still struggle to dynamically capture interactions between different feature categories, such as engagement levels, assessment performance, and demographic factors. For instance, a student who excels in tutor-marked assessments (TMA) while underperforms in computer-marked assessments (CMA) may face difficulties in objective assessment formats, impacting overall course performance. Traditional Transformer models process these features independently, failing to capture

interaction-driven performance patterns, which vary significantly across disciplines. For example, STEM courses often emphasize CMA, objective assessments, while humanities courses rely more on essay-based tutor-marked evaluations, leading to different student success factors across fields.

Despite advancements in ML and DL approaches, existing models still face two key challenges: (1) the inability to effectively capture and model complex, multi-dimensional relationships within aggregated student data, and (2) the lack of interpretability, which limits their real-world applicability in educational decision-making. This lack of transparency makes it difficult for educators to develop effective intervention strategies, as they cannot easily identify the key factors influencing student outcomes. To address these issues, a more adaptive, interpretable, and high-performance predictive model is required.

Despite these advancements, existing models fail to address the unique challenges of student performance prediction in structured datasets. Traditional ML models are limited by their inability to model complex, non-linear interactions, while tree-based models require extensive tuning and still struggle with high-dimensional feature spaces. Meanwhile, DL approaches, particularly Transformers, have demonstrated strong predictive capabilities in sequential data analysis but remain ineffective in structured student datasets where feature dependencies do not follow a temporal sequence. Although some hybrid ML-DL models attempt to mitigate these issues, they often lack interpretability, making it difficult for educators to understand the key drivers of student success. Furthermore, most existing models fail to dynamically capture cross-feature interactions between engagement behaviors, assessments, and demographics, leading to suboptimal prediction accuracy and limiting their applicability in real-world educational settings.

To bridge these gaps, a more adaptive, interpretable, and high-performance model is required—one that can effectively capture non-sequential feature dependencies, model high-dimensional relationships, and provide clear interpretability for educational decision-making. This study introduces KANFormer, an attention-enhanced deep learning model that integrates multi-head self-attention (MHSA), Feature-wise attention, and Kolmogorov-Arnold networks (KAN) to enhance student performance prediction. Unlike standard Transformers, which model sequential dependencies in time-series data, MHSA enables the model to learn relationships across different feature groups, regardless of order or temporal positioning. Traditional Transformers process input features independently and struggle to model complex cross-feature interactions in structured student data, where engagement, assessments, and demographics do not follow a strict sequence. MHSA overcomes this limitation by dynamically assigning importance to student behaviors, capturing engagement-assessment dependencies more effectively.

For instance, a student who performs well in quizzes while struggling in assignments may face different academic challenges than one who excels in coursework but underperforms in timed exams. Traditional Transformers may treat these attributes separately, whereas MHSA allows KANFormer to attend to multiple feature relationships simultaneously, improving its ability to identify nuanced performance patterns.

The Feature-Wise Attention layer improves interpretability by dynamically reweighting feature contributions, ensuring that direct academic assessments (e.g., assignments, exams) have a greater influence than indirect engagement metrics (e.g., forum activity), while still recognizing their complementary role. Meanwhile, KAN efficiently captures high-dimensional, non-linear dependencies, enabling more precise and interpretable predictions of student success without requiring explicitly designed feature interactions. By integrating these mechanisms, KANFormer bridges the gap between traditional ML models, which lack predictive power, and complex deep learning models, which lack interpretability. This novel approach offers a scalable, interpretable, and highly effective solution for improving student retention and success in online learning environments.

This study makes three key contributions:

- Introduces KANFormer, a novel attention-enhanced deep learning model for predicting student performance, handling both binary and multi-class outcomes to identify at-risk students.
- Evaluates KANFormer on a large-scale dataset (OULAD), demonstrating improvements over state-of-the-art models in integrating learning behaviors, assessments, and demographics.
- Provides interpretability analysis using SHAP values, explaining how different student behaviors influence predictions through multiple performance metrics such as accuracy, precision, recall, and AUC.

Outline of the paper Section 2 reviews related research. Section 3 describes the dataset used in experiments. Section 4 details the KANFormer model. Section 5 presents experimental results, while Sect. 6 introduces SHAP-based explainability. Finally, Sect. 7 concludes the paper and suggests future research directions.

2 Related works

Early research on student performance prediction mainly relied on traditional ML models, including decision trees (DT) (Wang et al. 2022), K-nearest neighbors (KNN) (Maghari 2018), logistic regression (LR) (Zhang et al. 2017), and support vector machines (SVM) (Al-Shehri et al.

2017). These models demonstrated moderate success in predicting student outcomes using structured features, such as assessment scores and demographics. However, LR and linear SVM assume linear relationships between features and target outcomes, limiting their ability to capture complex, non-linear dependencies and feature interactions, making them inadequate for modeling dynamic student engagement behaviors (Alnasyan et al. 2024; Aljaloud et al. 2022; Yang and Bai 2022).

In contrast, tree-based models (e.g., Decision Trees, Random Forests, Gradient Boosting) and KNN can effectively model non-linearity by modeling cross-feature dependencies in aggregated student data. However, these models still face limitations, tree-based models struggle with deep feature relationships without extensive ensemble techniques, whereas KNN's performance deteriorates in high-dimensional spaces due to the curse of dimensionality. Additionally, both approaches often require extensive hyperparameter tuning to optimize performance.

With advances in DL, models such as deep feedforward neural networks (DFFNNs) (Adnan et al. 2021) and convolutional neural networks (CNNs) (Poudyal et al. 2022) have demonstrated improved predictive accuracy in predicting student performance. More recent studies have explored long short-term memory (LSTM) networks and Transformer-based architectures to model sequential engagement data, effectively capturing learning patterns over time (Liu et al. 2022; Yang and Bai 2022; Zhang et al. 2023).

Despite their improved accuracy, several recent studies attempted to address these challenges while still exhibiting critical limitations. For example, the DFFNN model (Adnan et al. 2021) applied random forests (RF) and deep neural networks (DNNs) to predict student performance, primarily relying on engagement logs and assessment scores. However, these models failed to capture complex interactions between student behaviors, assessments, and demographic attributes, which are crucial for understanding student success. Similarly, Transformer-based architectures prioritize sequential dependencies, making them effective for time-series engagement data (e.g., clickstream logs). For example, Shou et al. (2024) applied LSTMs to engagement logs, achieving strong early predictions while struggling with generalizability across student profiles. Kusumawardani et al. (2023) focused on daily log activities, excluding assessment and demographic data, reducing predictive performance.

While these models improved upon traditional approaches, they remained constrained by limited feature sets, weak cross-feature modeling, and a lack of interpretability. Although Transformer-based models have demonstrated strong performance in sequential learning tasks, they are less suited for structured student datasets, where features are aggregated rather than strictly sequential. These models often lack explicit mechanisms to model

cross-feature interactions within multi-modal student data, limiting their ability to generalize effectively across different student profiles and learning environments (Yang and Bai 2022). Without specialized adaptations such as Feature-Wise Attention, these models struggle to effectively learn discipline-specific success factors. Additionally, DL models frequently function as black-box systems, limiting interpretability and restricting educators from deriving meaningful insights to support personalized interventions. Without explainability enhancements, such as SHapley Additive exPlanations (SHAP), understanding the contribution of engagement metrics, assessment patterns, and demographic factors to student performance remains challenging, reducing the practical applicability of these models in real-world educational settings.

While existing ML and DL models have shown promise in student performance prediction, they often struggle with capturing complex, multi-dimensional feature interactions and providing interpretability, limiting their practical use in education. To address these challenges, this study introduces KANFormer, a deep learning model that integrates multi-head self-attention (MHSA), feature-wise attention, and Kolmogorov–Arnold networks (KAN). KANFormer improves upon previous approaches by dynamically modeling engagement-assessment dependencies, prioritizing key predictors, and efficiently capturing high-dimensional relationships.

Unlike traditional ML and DL models that assume simplistic or sequential dependencies, KANFormer leverages MHSA to identify hidden patterns, such as how different combinations of quiz scores, discussion participation, and assignments influence student success. Additionally, feature-wise attention enhances interpretability by ensuring that direct academic assessments play a more prominent role in predictions than indirect engagement metrics, making it easier for educators to identify at-risk students and key success drivers.

Finally, KAN enhances predictive accuracy by learning complex dependencies without requiring manually defined feature interactions. This results in a robust model that generalizes well across different datasets, making KANFormer a scalable and effective solution for long-term student performance assessment and retention analysis (Table 1). In summary, KANFormer bridges the gap between prediction accuracy and explainability, providing educators with a powerful tool for early intervention and improved student outcomes in online learning environments.

3 Dataset

The open university learning analytics dataset (OULAD) is a widely used open-access dataset in educational data mining, containing student demographic data, academic performance records, and engagement metrics (Kuzilek et al. 2017). It includes 32,593 students across 22 courses from 2013 and 2014, covering STEM and social science disciplines. Structured into seven relational tables, it enables comprehensive analysis of student interactions within virtual learning environments (VLEs).

The dataset is categorized into four key feature groups:

- Demographic features, includes education level, prior attempts, and studied credits.
- Course information, captures module length and presentation schedules.
- Student activity features, tracks engagement with 20 VLE activity types (e.g., forums, quizzes, and course materials).
- Academic characteristics, includes final exam scores, computer-marked (CMA) and tutor-marked assessments (TMA), and assessment interactions.

The dataset's key features, descriptions, and data types are summarized in Table 2, which categorizes essential

Table 1 Comparison of existing models with KANFormer

Author (Year)	Feature integration	Handling non-linear data	Interpretability	Early prediction accuracy
Adnan et al. (2021)	Uses fixed feature sets (clickstreams, assessment scores)	Limited feature interaction modeling	Lacks explainability (uses Random Forest)	Improves prediction as course progresses
Kusumawardani et al. (2023)	Only uses log activities, ignoring assessments/demographics	No explicit feature interaction modeling	No interpretability for feature interactions	Predicts daily but lacks accuracy for final results
Shou et al. (2024)	Uses LSTMs but limited cross-feature interactions	Captures some relationships but lacks generalizability	Uses attention but doesn't explain feature interactions	Strong in early-stage prediction but may overfit
KANFormer (Ours)	Uses MHSA + KAN to learn cross-dimensional dependencies	KAN models complex feature dependencies dynamically	Feature-Wise Attention enhances interpretability	Combines early and final-stage prediction using multi-level attention

Table 2 Key features of the dataset

Feature category	Description	Data type
Demographic features	Includes highest education level, prior attempts, studied credits, and final outcomes, capturing essential student attributes	Categorical
Course information	Covers module length and presentation times (e.g., February “B” and October “J”), which influence course structure and deadlines	Categorical
Student activity features	Tracks engagement with the virtual learning environment (VLE), including forums, quizzes, and course materials	Numerical
Academic characteristics	Includes final exam scores, computer-marked (CMA) and tutor-marked assessments (TMA), along with interactions between assessment types	Numerical

elements used in the analysis. The demographic, academic, and engagement features were selected based on their relevance for predicting student performance.

Student performance is classified into Distinction (D), Pass (P), Fail (F), and Withdraw (W), supporting both binary and multi-class classification tasks. The dataset shows 53% of students were unsuccessful (22% failed, 31% withdrew), highlighting the need for targeted interventions to improve retention and success rates.

4 Methodology

This section outlines the dataset preprocessing and KANFormer model architecture (Fig. 1).

4.1 Data preprocessing

Data preprocessing involved data cleaning, merging, feature engineering, encoding, and grouping to prepare the dataset for training.

- *Cleaning & merging* Erroneous samples with inconsistent date fields were removed, and missing values were

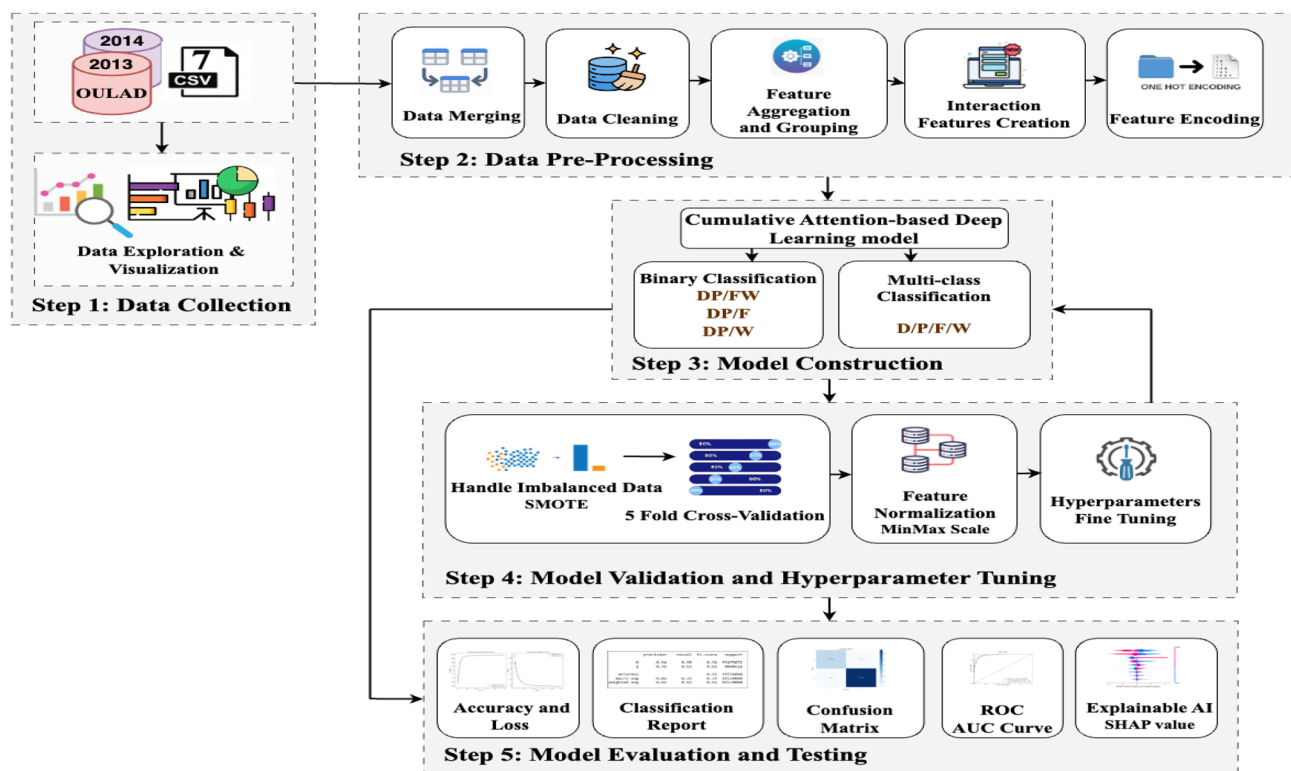


Fig. 1 The proposed attention-enhanced deep learning model (KANFormer) framework

imputed using mean imputation for numerical or mode imputation for categorical. Multiple CSV files (assessments, student interactions, and demographic data) were merged to provide a comprehensive student performance view.

- **Feature engineering** Several new features were created to enhance predictive power:
 - *Engagement ratio* measures total activity interactions relative to overall clicks.
 - *Completion rate* tracks student progress by comparing active days to course length.
 - *Interaction metrics* captures relationships between TMA and CMA scores.
 - *Peer comparison metrics* calculates average TMA, CMA, and Exam scores within each cohort for comparative analysis.
- **Encoding & grouping** Categorical variables (e.g., course module, education level) were one-hot encoded, and features were grouped into static (demographics, course info) and dynamic (academic performance, engagement, learning activities) to enable better pattern recognition in the model.

4.2 Feature selection & reduction

To refine the dataset, features with a Pearson correlation coefficient above 0.80 were removed (see Appendix A) to reduce multicollinearity and redundancy. A threshold of 0.80 was selected based on statistical best practices, as highly correlated features (≥ 0.80) tend to carry redundant information, which can lead to overfitting and reduced model interpretability. Retaining highly correlated features may cause the model to assign excessive weight to duplicate information rather than identifying distinct and meaningful patterns in the data. By removing such features, the model can focus on unique predictors, improving both generalizability and performance. Following the feature elimination process, the dataset was organized into static (long-term attributes) and dynamic (engagement-based features) groups to facilitate a structured learning process. A two-step correlation analysis was conducted to further refine feature selection. First, a between-group correlation analysis (Appendix 7) was performed to eliminate redundant relationships across different feature categories. Second, a within-group correlation analysis (Appendix 8) was applied to identify and remove strongly correlated attributes within each category, ensuring a more balanced feature representation. Finally, padding and masking techniques were implemented to ensure data compatibility with the attention-based model, standardizing input sequences while preventing artificial padding values from influencing the learning process. These steps collectively

enhance the robustness and interpretability of the dataset, allowing for more reliable and meaningful student performance predictions.

5 Model structure

The proposed attention-enhanced deep learning model (KANFormer) enhances student performance prediction by capturing feature interactions across engagement, assessment, and demographic factors. Unlike traditional models that process features independently, KANFormer integrates multi-head self-attention (MHSA), a feature-wise attention layer (FWAL), and Kolmogorov–Arnold networks (KAN) to learn non-linear dependencies effectively.

The model's input comprises features categorized into five key groups: course information, academic features, engagement features, activity features, and demographic features. Each feature vector is denoted as:

$$X = [F_1, F_2, F_3, \dots, F_n] \quad (1)$$

where $F_1, F_2, F_3, \dots, F_n$ are the individual features. These are normalized before processing. Static features (such as demographic attributes) are passed directly into the model, while dynamic features (such as engagement and academic performance) undergo attention-based transformation through MHSA.

5.1 Multi-head self-attention (MHSA)

The multi-head self-attention (MHSA) mechanism evaluates the importance of different feature groups by dynamically weighting their contributions. For example, a student who spends more time reviewing video lectures may have a different success trajectory compared to one who primarily engages in quizzes. MHSA ensures that the model assigns different attention weights to engagement types based on their impact on academic success.

Mathematically, MHSA transforms input features into Query (Q), Key (K), and Value (V) matrices, where the attention scores are computed using the scaled dot-product attention mechanism:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V \quad (2)$$

where: - $Q = XW_Q$, $K = XW_K$, and $V = XW_V$ are the query, key, and value matrices, - d_k is the dimensionality of the key vectors, used for scaling.

To further enhance performance, Multi-Head Attention (MHA) extends this mechanism across multiple attention

heads, each capturing different aspects of the feature relationships:

$$\text{MHSA}(X) = \text{Concat}(\text{head}_1, \dots, \text{head}_h)W_O \quad (3)$$

where W_O is the output projection matrix, and each head applies Eq. 2 independently.

This process allows the model to prioritize meaningful interactions, recognizing that quizzes and assignments contribute differently to student success depending on course difficulty.

5.2 Feature-wise attention layer (FWAL)

FWAL improves interpretability by adaptively reweighting feature contributions. Instead of treating all features equally, FWAL dynamically prioritizes critical academic assessments while still considering complementary engagement indicators. The attention scores are computed as:

$$\alpha = \text{softmax}(WF + b) \quad (4)$$

where:

- $W \in \mathbb{R}^{d \times d}$ is a learnable weight matrix,
- $b \in \mathbb{R}^d$ is the bias vector,
- α is the attention score vector, determining the importance of each feature.

The final feature representation is computed as:

$$F_{\text{final}} = \alpha \odot F \quad (5)$$

where $\odot$ denotes element-wise multiplication.

This mechanism ensures that critical academic assessments (e.g., exam scores, assignments) are given higher importance compared to indirect engagement metrics (e.g., forum activity), while still considering their complementary role. This enhances model interpretability, enabling educators to identify at-risk students and key success factors more effectively.

5.3 Kolmogorov–Arnold network (KAN)

The KAN enhances predictive accuracy by capturing high-dimensional, non-linear dependencies in student data. Unlike conventional neural networks, which require manually defined feature interactions, KAN automatically learns complex relationships, improving model robustness.

KAN is based on the Kolmogorov–Arnold representation theorem, which states that any multivariate function can be expressed as:

$$f(x_1, x_2, \dots, x_n) = \sum_{q=1}^Q \varphi_q \left(\sum_{p=1}^P \psi_{pq}(x_p) \right) \quad (6)$$

where: - φ_q and ψ_{pq} are learnable functions modeled via neural networks, - Q and P control the decomposition complexity.

Unlike traditional neural networks that process each feature independently, KAN explicitly models feature dependencies, enabling more hidden predictions. For instance, it can recognize that students with high quiz scores and low discussion participation tend to excel in exams, whereas those who rely heavily on discussions may struggle with standardized tests.

By incorporating KAN into KANFormer, the model:

- Eliminates the need for predefined feature interactions,
- Learns high-dimensional dependencies automatically,
- Generalizes well across different datasets, making it applicable for long-term performance assessment and student retention analysis.

5.4 Summary of KANFormer advantages

By integrating MHSA for dynamic feature weighting, FWAL for interpretability, and KAN for non-linear modeling, KANFormer provides a scalable and robust solution for student performance prediction. The overall architecture is illustrated in Fig. 2.

6 Experimental

This section outlines the class imbalance handling, cross-validation, hyperparameter tuning, evaluation metrics, and computational environment used in the experiments.

6.1 Handling class imbalance

The final result distribution of the 32,593 students has shown a significant class imbalance, with instances of “Distinction” (3024), “Pass” (12,361), “Fail” (7052), and “Withdrawn” (10,156). The imbalance problem affects the performance of predictive models. The imbalance problem affects the performance of predictive models, which can be fixed by using techniques at the data (preprocessing) or algorithm (algorithm-level) level (Chawla et al. 2002; Werner de Vargas et al. 2023; Flores et al. 2022). To address this, SMOTE (synthetic minority over-sampling technique) was applied to up-sample the minority classes (“Distinction” and “Fail”) by generating synthetic examples through K-nearest neighbor interpolation (Skowron and Wojna 2004; Wang et al. 2023). Despite all the

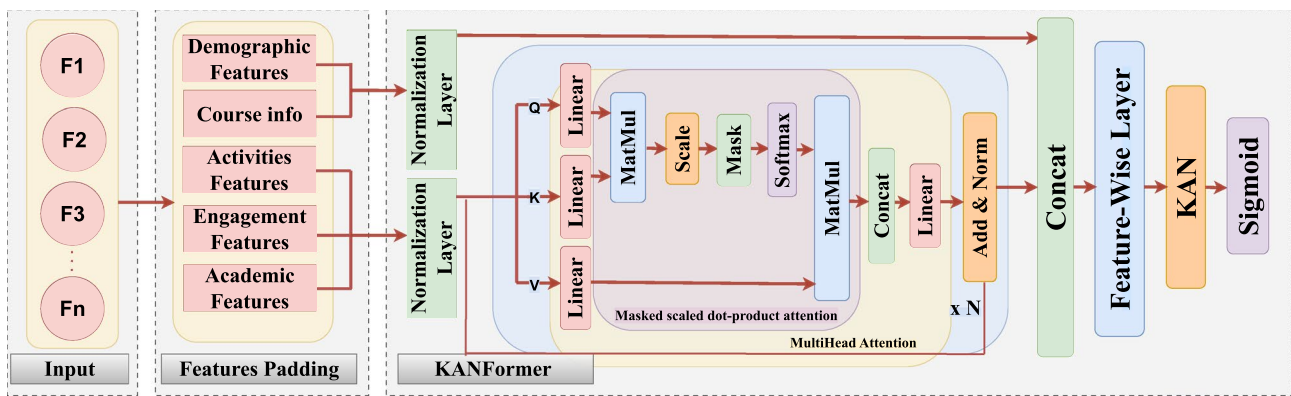


Fig. 2 The proposed attention-enhanced deep learning (KANFormer) architecture

Table 3 Hyper-parameter settings

Parameters name	Hyper-parameter settings
Learning rate	0.0001
Multi-head attention blocks	2
Attention heads	2
Kolmogorov–Arnold network	Hidden layer with 32 neurons Grid size: 5 Spline order: 3
Dropout rate	50%
Batch size	64
Loss function	Cross-entropy
Optimizer	AdamW
Training epochs	50

benefits of SMOTE, it has limitations, including the potential to lead to overfitting, ignore within-class imbalance, and increase noise (Wang et al. 2023; Werner de Vargas et al. 2023).

6.2 Cross-validation

A 5-fold cross-validation approach was used to evaluate model performance by splitting the dataset into five equal subsets, where each subset served as the test set once while the remaining four were used for training. This method

Table 4 Confusion matrix

	Predicted positive	Predicted negative
Actual positive	TP	FN
Actual negative	FP	TN

ensures robust generalization, reduces overfitting, and is particularly crucial when handling imbalanced datasets with SMOTE to prevent synthetic data bias (Zhang and Liu 2023). Additionally, MinMax scaling was applied within each fold, normalizing features between 0 and 1. The scaler was first fitted on the training set and then applied to the validation set to prevent data leakage, ensuring balanced and consistent feature distributions during training.

6.3 Hyperparameter tuning and model configuration

This study employs Optuna for Bayesian optimization using the tree-structured parzen estimator (TPE), which iteratively refines hyperparameter selection to enhance model performance (Akiba et al. 2019). Key parameters tuned include attention heads, embedding dimensions, dropout rate, learning rate, batch size, and spline order.

To improve efficiency, early stopping halts training after 10 epochs of no validation improvement, and a Learning Rate Scheduler (ReduceLROnPlateau) dynamically adjusts the learning rate for faster convergence while avoiding local minima. The final KANFormer model settings are summarized in Table 3.

6.4 Evaluation indicators

The performance of the KANFormer model is evaluated using accuracy, precision, recall, F1-score, ROC AUC, and a confusion matrix. These metrics assess the model's effectiveness, particularly in handling class imbalance.

Accuracy measures the proportion of correctly classified instances:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (7)$$

Table 5 KANFormer results for three different configurations (DP/FW, DP/F, DP/W)

KANFormer	DP/FW			DP/F			DP/W					
	D/P	F/W	Macro avg.	Weighted avg.	D/P	F	Macro avg.	Weighted avg.	D/P	W	Macro avg.	Weighted avg.
Precision	0.9215	0.9711	0.9494	0.9495	0.9122	0.9573	0.9347	0.9346	0.9901	0.9939	0.9920	0.9920
Recall	0.9729	0.9166	0.9482	0.9482	0.9600	0.9066	0.9333	0.9334	0.9940	0.9898	0.9919	0.9920
F1-score	0.9431	0.9465	0.9481	0.9481	0.9355	0.9313	0.9334	0.9334	0.9921	0.9918	0.9919	0.9920
Average accuracy	0.9459				0.9361				0.9921			
95% confidence interval	0.9438 - 0.9487				0.9316 - 0.9431				0.9901 - 0.9936			
Average loss	0.1453				0.1624				0.0252			
Standard deviation	± 0.0021				± 0.0047				± 0.0012			
Average ROC AUC	0.9835				0.9809				0.9992			

Precision indicates how many of the predicted positive cases are actually correct:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (8)$$

Recall measures the proportion of actual positives correctly identified:

$$\text{Recall} = \frac{TP}{TP + FN} \quad (9)$$

F1-score provides a balanced measure of precision and recall:

$$\text{F1-Score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (10)$$

The ROC AUC (receiver operating characteristic - area under curve) evaluates the model's ability to distinguish between positive and negative classes. It plots the true positive rate (TPR) against the false positive rate (FPR) at various classification thresholds, where:

$$\text{TPR} = \frac{TP}{TP + FN}, \text{FPR} = \frac{FP}{FP + TN}$$

An AUC score of 0.5 indicates random guessing, while 1.0 represents perfect classification.

The confusion matrix (Table 4) provides a breakdown of the model predictions for actual positive and negative cases.

6.5 Experimental environment

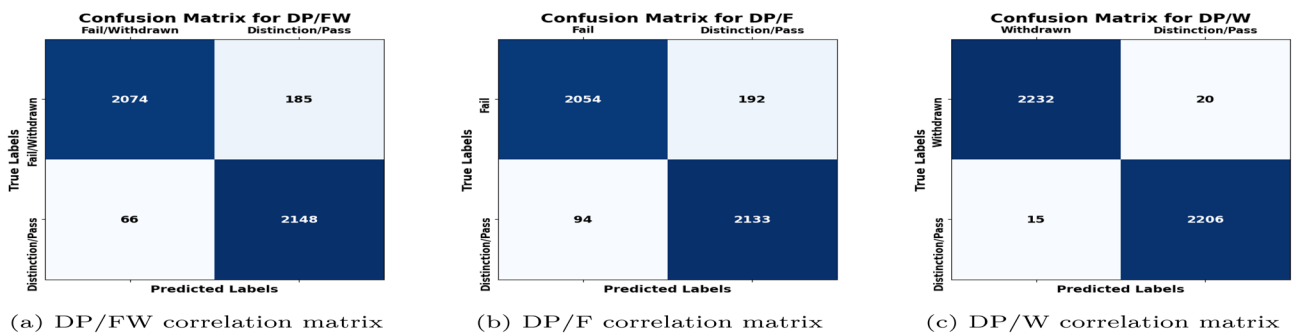
The experiments were conducted on Google Colab, using NVIDIA T4 GPUs to accelerate model training and efficiently handle large datasets. The Python 3.10 programming language was used, with TensorFlow 2.15.0 and Keras for model development and training. Additionally, Scikit-learn 1.3.2 was employed for data preprocessing, including feature scaling, cross-validation, and performance evaluation. This cloud-based setup ensured efficient execution of DL tasks while optimizing computational resources.

6.6 Experimental results and analysis

The KANFormer model is evaluated for its effectiveness in predicting student performance across binary and multi-class classification tasks. In the binary classification scenario, the model predicts whether a student belongs to the successful category ("Distinction" and "Pass") or the unsuccessful category ("Fail" and "Withdrawn"). Additionally, separate evaluations are conducted to assess the model's ability to distinguish ("Distinction" and "Pass") from "Fail" and ("Distinction" and "Pass") from "Withdrawn" individually. In the multi-class classification scenario, the model predicts

Table 6 Comparison of binary-class classification prediction performance across different DL models

	DP/FW		DP/F		DP/W	
	Accuracy	F1-score	Accuracy	F1-score	Accuracy	F1-score
Decision tree	0.9173	0.9173	0.9063	0.9063	0.9898	0.9898
KNeighbors	0.9235	0.9234	0.9054	0.9053	0.9836	0.9836
Logistic regression	0.9361	0.9361	0.8974	0.8972	0.9898	0.9898
TEnc	0.9357	0.9321	0.9175	0.8627	0.9814	0.9710
Kusumawardani et al. (2023)						
MTAPSP	0.9432	0.9433	0.9271	0.9254	0.9892	0.9892
Shou et al. (2024)						
KANFormer	0.9459	0.9454	0.9361	0.9341	0.9921	0.9920

**Fig. 3** The confusion matrices for DP/FW, DP/F, and DP/W tasks

all four outcome categories: “Distinction,” “Pass,” “Fail,” and “Withdrawn”.

To understand the contribution of individual components, ablation experiments are conducted, analyzing the roles of MHSA and KANs within the model. Finally, to assess generalization and robustness, the KANFormer model is tested beyond the OULAD dataset, incorporating three additional datasets: ASSISTments2012, xAPI, and the Mathematics and Portuguese Language Courses dataset. This broader evaluation ensures the model’s adaptability across different educational contexts and datasets.

6.6.1 Binary classification prediction performance experiments

The KANFormer model is evaluated across binary classification tasks to assess its effectiveness in distinguishing student performance categories. These evaluations provide a comprehensive measure of the model’s reliability in predicting academic outcomes. As summarized in Table 5, KANFormer achieves strong performance across all classification groups, with particularly high accuracy in distinguishing Distinction/Pass (DP) vs. Withdrawn (W). This classification task achieves near-perfect precision, recall, F1-score, and

ROC AUC, highlighting the model’s robustness in identifying students at-risk of withdrawal. The high precision and accuracy in DP/W classification indicate that KANFormer effectively captures student learning behavior, assessment patterns, and demographic trends, enabling more reliable predictions of student withdrawal.

While DP/F (Distinction/Pass vs. Fail) and DP/FW (Distinction/Pass vs. Fail/Withdrawn) tasks exhibit slightly lower performance, the model maintains high accuracy and reliability across all settings. The DP/W task emerges as the most stable and precise classification, as reflected in the lowest average loss, minimal standard deviation, and highest accuracy and ROC AUC scores. The classification of pass/fail students proves more challenging due to overlapping engagement patterns and assessment scores, making the distinction between these groups less clear. Overall, these results demonstrate the effectiveness of KANFormer in accurately distinguishing student performance categories, making it a reliable predictive framework for identifying at-risk students and supporting early intervention strategies in online education.

The evaluation results demonstrate that KANFormer achieves high accuracy across all classification tasks, with particularly strong performance in distinguishing

Table 7 Comparison of multi-class classification prediction performance across different ML and DL models

	Decision tree	KNeighbors	Logistic regression	DFNN (Adnan et al. 2021)	MTAPSP (Shou et al. 2024)	KANFormer
<i>Precision</i>						
Distinction	0.85	0.79	0.72	0.66	0.58	0.86
Pass	0.77	0.72	0.64	0.74	0.79	0.81
Fail	0.60	0.63	0.65	0.55	0.62	0.69
Withdrawn	0.70	0.69	0.71	0.76	0.78	0.73
Macro avg.	0.73	0.71	0.68	0.68	0.69	0.77
Weighted avg.	0.73	0.71	0.68	0.70	0.73	0.77
<i>Recall</i>						
Distinction	0.85	0.84	0.77	0.47	0.40	0.89
Pass	0.76	0.77	0.65	0.90	0.89	0.81
Fail	0.61	0.50	0.54	0.36	0.51	0.60
Withdrawn	0.70	0.75	0.78	0.80	0.83	0.76
Macro avg.	0.73	0.71	0.68	0.63	0.66	0.76
Weighted avg.	0.73	0.71	0.68	0.71	0.74	0.77
<i>F1-Score</i>						
Distinction	0.85	0.81	0.75	0.55	0.47	0.87
Pass	0.76	0.74	0.64	0.81	0.84	0.81
Fail	0.61	0.56	0.59	0.44	0.56	0.64
Withdrawn	0.70	0.72	0.74	0.78	0.80	0.75
Macro avg.	0.73	0.71	0.68	0.64	0.67	0.77
Weighted avg.	0.73	0.71	0.68	0.70	0.73	0.77
Accuracy	0.73	0.71	0.68	0.71	0.74	0.77

The bold values in the indicate the best-performing scores for each metric across the different models

Distinction/Pass (DP) from Withdrawn (W) students, achieving an accuracy of 99.21%. This suggests that withdrawn students exhibit distinct engagement patterns, making them easier to identify compared to other student categories. However, the model exhibits lower accuracy in the DP vs. Fail (DP/F) classification task (93.61%), indicating that differentiating borderline pass and fail students is more challenging due to overlapping behavioral patterns. Further analysis of recall values highlights the model's effectiveness in early identification of at-risk students, particularly in predicting withdrawal risk, where it achieves over 96% recall. This suggests that KANFormer can be a valuable tool for early intervention strategies, helping educators detect disengaged students before they drop out. However, the model's lower recall in the DP/F task indicates potential difficulties in distinguishing students who are on the verge of failing, emphasizing the need to incorporate additional engagement-based predictors to improve classification performance in borderline cases.

Five benchmark models, decision tree classifier (DT), Kneighbors classifier (KN), logistic regression (LR),

transformer encode model (TEnc) (Kusumawardani et al. 2023). Multi-dimensional time-series analysis-based Student performance prediction model (MTAPSP) (Shou et al. 2024), are used for comparative experiments to validate the advantages of the KANFormer model for binary classification prediction, using the student click-stream data, their assessment scores, and the data of the student demographic characteristics. Each of these model performances is shown in Table 6.

The binary classification prediction experiments in Table 6 show that the KANFormer model performs well in all binary classification tasks (DP/FW, DP/F, and DP/W). It performs better than the TEnc and MTAPSP models in terms of accuracy and F1 score, showing that it can handle different performance levels. Teachers can predict student performance using the KanFormer model, gain insight into the academic level of their students, and develop targeted instructional programs accordingly. For instance, the KANFormer model recommends higher-level courses to students predicted to have "Distinction" results, aiming to deepen their learning and enhance their abilities. In addition, it

provides the necessary support and assistance for students expected to leave school, addressing any academic difficulties they may encounter. The application of the KANFormer model offers an effective tool for predicting student performance. Provides a powerful means of guidance for educators, allowing them to counsel students in a more comprehensive and individualized manner and promoting the improvement of teaching quality.

The confusion matrices illustrate the KANFormer model's performance across binary classification tasks: DP/FW, DP/F, and DP/W, as shown in Fig. 3. Each matrix provides insights into the model's predictive accuracy by displaying correctly classified instances along the diagonal. Across all tasks, the model exhibits strong predictive capability, particularly in the DP/W classification, where it achieves near-perfect classification of distinction and withdrawn students. This high performance suggests that withdrawn students exhibit distinct engagement patterns, making them easier to separate from those who successfully complete their courses. In the DP/F task, distinguishing between distinction/pass and fail categories presents a greater challenge. The model demonstrates slightly lower performance, as evidenced by an increase in false positives and false negatives, indicating occasional misclassifications low-performing students

as passing or distinction. This challenge may arise because students on the pass/fail boundary share similar engagement behaviors and assessment scores, making their classification less distinct. For the DP/FW classification, the model maintains a well-balanced accuracy, with minimal misclassification between the distinction/pass and fail/withdraw groups. The model effectively identifies students at distinction and withdraw of performance, with minor errors occurring within pass and fail categories due to the overlap in learning behavior between failing and withdrawn students.

Overall, the KANFormer model achieves high true positive and true negative rates across all tasks, reinforcing its robustness in predicting student outcomes. The strongest performance is observed in DP/W, followed by DP/FW, with DP/F presenting the greatest classification challenge. These findings suggest that, while the model excels at distinguishing students with clear success or failure trajectories, improving its ability to learn nuanced patterns in borderline cases, particularly between pass and fail, would improve predictive accuracy. The results further indicate that integrating MHSA and leveraging deep learning techniques to capture complex nonlinear relationships in student engagement and academic performance significantly improve classification performance.

6.7 Multi-classification prediction performance

The KANFormer model is evaluated for its effectiveness in multi-class classification tasks. To assess its performance, five ML and DL models are employed for comparison: decision tree classifier (DT), K-nearest neighbors classifier (KN), logistic regression (LR), deep feedforward neural network (DFFNN) (Adnan et al. 2021), and the multi-dimensional time-series analysis-based student performance prediction model (MTAPSP) (Shou et al. 2024). The comparative results of these models are presented in Table 7.

The evaluation results indicate that KANFormer performs best in predicting Distinction students, achieving an F1-score of 87%. This strong performance is likely attributed to consistent engagement patterns and high assessment scores, which provide clear indicators of academic success. In contrast, the Fail category had the lowest recall (60%),

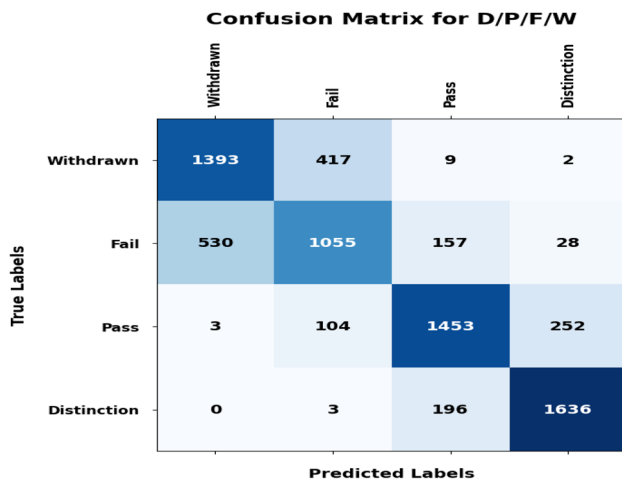


Fig. 4 The confusion matrix for D/P/F/W task

Table 8 Results of ablation experiments applied for DP/FW task

Model	Precision	Recall	F1-score	Loss	Average accuracy	Average ROC AUC	95% confidence interval for accuracy	Standard deviation
Standalone KAN	0.9409	0.9391	0.9390	0.1569	0.9391	0.9814	[0.9363, 0.9423]	± 0.0020
Standalone Former	0.9388	0.9366	0.9365	0.1667	0.9366	0.9782	[0.9341, 0.9411]	± 0.0027
KANFormer without FWAL	0.9334	0.9286	0.9284	0.1761	0.9286	0.9773	[0.9220, 0.9329]	± 0.0039
KANFormer	0.9495	0.9482	0.9481	0.1453	0.9459	0.9835	[0.9438, 0.9487]	± 0.0021

suggesting that students at risk of failing exhibit behaviors that overlap with those in the Pass category, leading to higher misclassification rates. Additionally, the model demonstrates strong predictability for Withdrawn students, achieving an F1-score of 75%, which suggests that students who disengage from the course exhibit distinct behavioral patterns that the model effectively captures. However, the challenges in differentiating Fail and Pass students indicate that assessment-based features alone may not be sufficient for accurate classification. Incorporating additional engagement metrics, such as forum participation trends and interaction levels, could improve the model's ability to identify borderline students who are at risk of failing. These findings emphasize the need for a comprehensive predictive approach that integrates both academic performance indicators and engagement behaviors to enhance early risk detection in online learning environments.

Table 7 shows that the KANFormer model outperformed most of the evaluation metrics. Although the MTAPSP model performed relatively well in several categories, especially in identifying students in the "Withdrawn" category, when considering the class imbalance and the importance of a balanced performance across metrics, KANFormer emerges as the superior model.

Specifically, KANFormer demonstrated the highest accuracy and outperformed all other models in the macro average and weighted average across the metrics of precision, recall, and F1-score. It excelled, particularly in the Distinction and Fail categories. For the Distinction category, KANFormer achieved a precision of 0.86, a recall of 0.89, and an F1-Score of 0.87, significantly outperforming all other models. Similarly, in the Fail category, KANFormer obtained a precision of 0.69, a recall of 0.60, and an F1-Score of 0.64, again showing its strong ability to identify high-achieving and at-risk students correctly.

Compared to other DL models such as DFFNN and MTAPSP, KanFormer demonstrated superior results in all metrics. This indicates that KANFormer captures complex patterns and relationships between learning activities and performance categories more effectively. The consistent

superiority of KANFormer in all these metrics suggests that it can handle data imbalance and is better at integrating various student features to predict results more accurately.

These results demonstrate the excellent performance of KANFormer in predicting student performance. Using KANFormer, educators can gain deeper insights into student academic performance and provide more personalized support. For example, educators can guide students who are predicted to achieve distinction toward more advanced learning opportunities while identifying students who are predicted to fail or withdraw, which can aid in their improvement. In general, the KANFormer model offers a powerful tool for predicting student performance, helping to enhance the quality of education by enabling more individualized support for students.

The confusion matrix in Fig. 4 highlights the ability of the KANFormer model to classify four categories, the diagonal line being the number of correctly predicted categories. The model performs exceptionally well in identifying distinction students, likely due to distinctive engagement patterns and strong academic performance. However, it exhibits challenges in differentiating fail and withdrawn students, as evidenced by a high misclassification rate between these two categories. This overlap suggests that students in these groups share similar behavioral and academic characteristics, making the classification more complex. The moderate misclassification of pass students indicates some similarities in features with the distinction category, likely due to marginal performance differences.

6.8 Ablation experiments

The ablation study evaluates the contribution of MHSA, feature-wise attention layer (FWAL), and KAN to the predictive performance of KANFormer, with results presented in Table 8. The findings indicate that removing any of these components leads to performance degradation, reinforcing their critical role in student performance prediction. MHSA plays a key role in capturing feature dependencies,

Table 9 Performance metrics of the KANFormer model across multiple educational datasets

Dataset	Average precision	Average recall	Average F1-score	Average accuracy	Average loss	Average ROC AUC	Standard deviation
ASESTEments2012	0.8345	0.7472	0.7787	0.8766	0.2904	0.9029	± 0.0029
xAPI	0.8450	0.8375	0.8199	0.8375	0.5128	0.9421	± 0.0193
Mathematics & Portuguese	0.8234	0.8008	0.7874	0.8008	0.4245	0.8851	± 0.0216
OULAD	0.9495	0.9482	0.9481	0.9459	0.1458	0.9835	± 0.0021

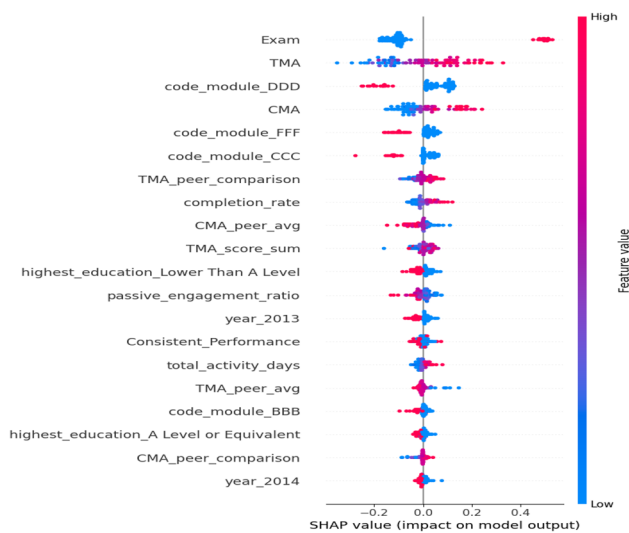


Fig. 5 SHAP Plot showing feature importance and the impact of features on model predictions

dynamically weighting engagement and assessment features. The Standalone KAN model (without MHSA) exhibits a lower accuracy of 93.91% and ROC AUC of 0.9814, compared to 94.59% and 0.9835 in the full KANFormer model. This decline suggests that removing MHSA weakens the model's ability to capture engagement-assessment relationships, reducing its ability to distinguish between high- and low-performing students.

Similarly, KAN significantly enhances non-linear feature interactions, improving the model's ability to learn complex patterns in student data. The Standalone Former model (without KAN) achieves a lower accuracy of 93.66% and ROC AUC of 0.9782, with a decline in F1-score from 0.9481 (KANFormer) to 0.9365. This confirms that KAN enhances feature representation by capturing intricate dependencies between student attributes, thereby improving classification performance. Additionally, the removal of FWAL results in the most noticeable performance decline, with accuracy dropping to 92.86%, F1-score decreasing to 0.9284, and loss increasing from 0.1453 to 0.1761. These findings highlight the importance of FWAL in prioritizing key predictive features, ensuring that essential academic and engagement attributes receive appropriate weight in decision-making. Without FWAL, the model struggles to distinguish feature importance, leading to higher classification errors.

6.9 Performance evaluation across multiple datasets

The study extended the evaluation to three additional datasets, ASSISTments2012, xAPI, and the Mathematics and Portuguese Language Courses dataset, to assess the generalizability and robustness of the KANFormer model across different educational contexts.

- ASSISTments dataset 2012-2013** at <https://sites.google.com/site/assistmentsdata/datasets/2012-13-school-data-with-affect>: The ASSISTments platform is widely used for knowledge-tracing tasks in mathematics education. ASSIST12, a sub-dataset collected during 2012-2013, offers one of the largest publicly available datasets with over 2,541,201 interactions, 27,066 students, and 45,716 questions. It includes detailed information such as student IDs, knowledge concept IDs, correctness of answers, and precise timestamps for each question. This granularity allows for an in-depth analysis of time-dependent factors, such as the intervals between questions and their impact on knowledge retention.
- The xAPI-Edu-Data dataset** at <https://www.kaggle.com/datasets/aljarah/xAPI-Edu-Data>: The xAPI-Edu-Data dataset, sourced from Alibaba Cloud Tianchi, comprises 480 student records collected over two semesters. The dataset includes 17 attributes related to student performance, such as education level, class, selected courses, frequency of raising hands in class, attendance characteristics, and parental information. Student grades are classified into three categories: low represented as "L" (< 60), medium represented as "M" ($60 \leq 90$), and high represented as "H" (≥ 90).
- the Mathematics and Portuguese Language Courses dataset** at <https://www.kaggle.com/datasets/mrigaankjaswal/student-performance-in-mathematics-and-portuguese/data>: The dataset from these two secondary school classes comprises roughly 1,044 students overall. There are 395 entries in the mathematics course and 649 records in the Portuguese language course dataset, each with 33 features. A wide range of characteristics, including demographic information (e.g., age, sex), social factors (e.g., parental education, family size), and academic performance metrics (e.g., study time, number of absences from school), were gathered through school reports and questionnaires. Although grades for three periods (G1, G2, and G3) are included in the dataset, G3 is chosen as the target variable for prediction since

it thoroughly evaluates student's academic performance and represents their cumulative success.

These datasets, representing various learning environments and student behaviors, complement the OULAD dataset used earlier. The KANFormer model's evaluation on multiple datasets aims to validate its performance consistency, adaptability, and effectiveness in predicting student outcomes across varied educational settings. The evaluation focused on a binary classification task, distinguishing between "Distinction/Pass" and "Fail/Withdrawn". All datasets were preprocessed to ensure fair and consistent evaluation, including handling missing values and feature scaling where necessary. Additionally, class imbalance issues were addressed using the Synthetic Minority Oversampling Technique (SMOTE), ensuring balanced class distributions and more reliable performance metrics. The results, summarized in Table 9, demonstrate the model's strong predictive performance and adaptability across various datasets, showcasing significant generalization capabilities.

On the ASSISTEments2012-2013 dataset, which lacks demographic features, the model maintained strong performance with an accuracy of 87.66% and a ROC AUC of 90.29%, highlighting its resilience when key features are unavailable. The xAPI dataset, despite its small sample size of 480 rows, demonstrated the model's ability to handle overfitting challenges, achieving an accuracy of 83.75% and a ROC AUC of 94.21%. For the Mathematics and Portuguese Language Courses dataset, which represents traditional classroom-based education that shows significant differences in feature distributions, the model achieved an accuracy of 80.08% and a ROC AUC of 88.51%. While this dataset posed additional challenges, the results underline the model's capacity to adapt to distinct academic contexts. These evaluations reflect the model's robust generalization across datasets with varying features, sample sizes, and domains, supporting its applicability to different educational contexts and its potential to predict student performance effectively in real-world scenarios.

7 Evaluation & explainable AI

To improve interpretability, we use SHAP (SHapley Additive Explanations) values to quantify each feature's contribution to predictions. SHAP allows to identify which student attributes directly impact learning outcomes, making

it easier for educators to design effective interventions. As shown in Fig. 5, red points represent high feature values, while blue points indicate low values. Positive SHAP values suggest a reduced failure risk, while negative values indicate underperformance risks.

SHAP analysis confirms that academic assessments, particularly exam scores, TMA, and CMA, have the highest impact on student performance predictions. Unlike engagement-based features, which provide supplementary learning insights, assessment scores remain the primary drivers of success. Interestingly, students who perform well in quizzes but engage less in discussions still tend to excel in exams, reinforcing the idea that forum participation is not always an indicator of academic achievement. Furthermore, SHAP results indicate that STEM modules (CCC, DDD, FFF) exhibit higher failure and dropout rates compared to non-STEM courses, suggesting that increased academic rigor influences student retention. These findings underscore the need for targeted interventions, where educators should prioritize assessment-based support while also recognizing engagement patterns as complementary, rather than primary, success indicators.

7.1 Comparing engagement metrics vs. academic assessments

The comparison between engagement metrics and academic assessments highlights that direct performance indicators remain the strongest predictors of student success, as confirmed by SHAP analysis. SHAP results demonstrate that exams, tutor-marked assignments (TMA), and computer-marked assignments (CMA) have the highest predictive importance, indicating that assessment performance plays a crucial role in determining student outcomes. In contrast, engagement-based factors such as forum participation and lecture viewing time receive lower SHAP values, suggesting that while engagement contributes to learning, it does not independently determine success as effectively as direct assessments.

While engagement provides useful learning insights, its impact on academic performance is context-dependent. SHAP results indicate that while students who actively participate in forums and lectures may enhance their learning experience, this does not consistently translate to higher grades. Additionally, the role of engagement varies by subject. In STEM courses, where performance is assessed through exams and problem-solving, assessment scores remain the most reliable indicators of success. Conversely,

in discussion-heavy humanities courses, engagement may play a more significant role in shaping learning outcomes.

7.2 Implications for educational interventions

The insights from SHAP analysis emphasize the need for data-driven intervention strategies in higher education. Since assessment scores are the strongest predictors of student success, early intervention should focus on students struggling in exams and TMAs. Based on these findings, institutions can implement:

- Real-time performance tracking in learning management systems (LMS): by integrating FWAL outputs into LMS dashboards, educators can receive automated alerts when students exhibit declining assessment performance.
- Adaptive coursework and personalized learning paths: students struggling in STEM modules (e.g., DDD, FFF) can be assigned adaptive exercises based on their predicted performance trends.
- Proactive engagement monitoring: while engagement features have lower predictive power, early identification of disengaged students can help universities offer personalized support before assessment failure occurs.

The combination of KANFormer and SHAP explainability provides a powerful framework for student performance prediction, offering both high predictive accuracy and interpretability. By leveraging these insights, institutions can enhance student retention through adaptive, data-driven interventions, ensuring at-risk students receive timely support.

8 Conclusion and future work

This study presents KANFormer, a novel deep learning model for student performance prediction, which introduces three key innovations over existing baseline models. Firstly, it employs feature grouping, padding, and masking techniques to optimize computational efficiency and focus on relevant data. Secondly, it integrates a hybrid architecture combining attention-enhanced mechanisms with Kolmogorov–Arnold networks (KANs), significantly enhancing predictive performance by capturing complex feature

interactions. Thirdly, it incorporates a feature-wise attention layer that dynamically adjusts the importance of different features, addressing the limitations of traditional models that treat all inputs equally. These advancements contribute to a more interpretable, efficient, and robust predictive framework for identifying at-risk students in VLEs.

Empirical evaluations demonstrate that KANFormer outperforms other models in binary and multi-class classification tasks. The model consistently achieves higher precision, recall, F1-score, and overall accuracy, highlighting its ability to differentiate between students at varying risk levels. Additionally, KANFormer exhibits strong stability and generalizability, as evidenced by its low loss values and narrow confidence intervals across multiple datasets. These findings indicate that the proposed model can serve as a reliable predictive tool for identifying at-risk students, enabling educational institutions to implement targeted support strategies.

Despite its effectiveness, this study relies on retrospective analysis, which, while valuable for identifying at-risk students in future courses, limits real-time intervention opportunities during an active learning period. Most existing predictive models operate at fixed time intervals (e.g., monthly, weekly or daily updates), which may fail to capture short-term behavioral fluctuations that signal an imminent risk of disengagement or failure. Addressing this limitation requires the development of real-time student performance tracking mechanisms capable of dynamically analyzing online interactions and adjusting predictions accordingly.

Future research should focus on extending KANFormer's capabilities in three key areas. Firstly, real-time performance tracking should be explored to enable continuous monitoring of student engagement, allowing for immediate intervention when risk patterns emerge. Secondly, personalized adaptive learning models should be developed, leveraging KANFormer's predictive power to recommend adapted learning pathways that respond to individual student needs. Finally, integrating KANFormer into learning management systems (LMS) would facilitate automated alerts and recommendations, enhancing educator's ability to provide timely support. Advancing in these directions will enable more proactive, data-driven educational interventions, ultimately improving student retention and overall academic success.

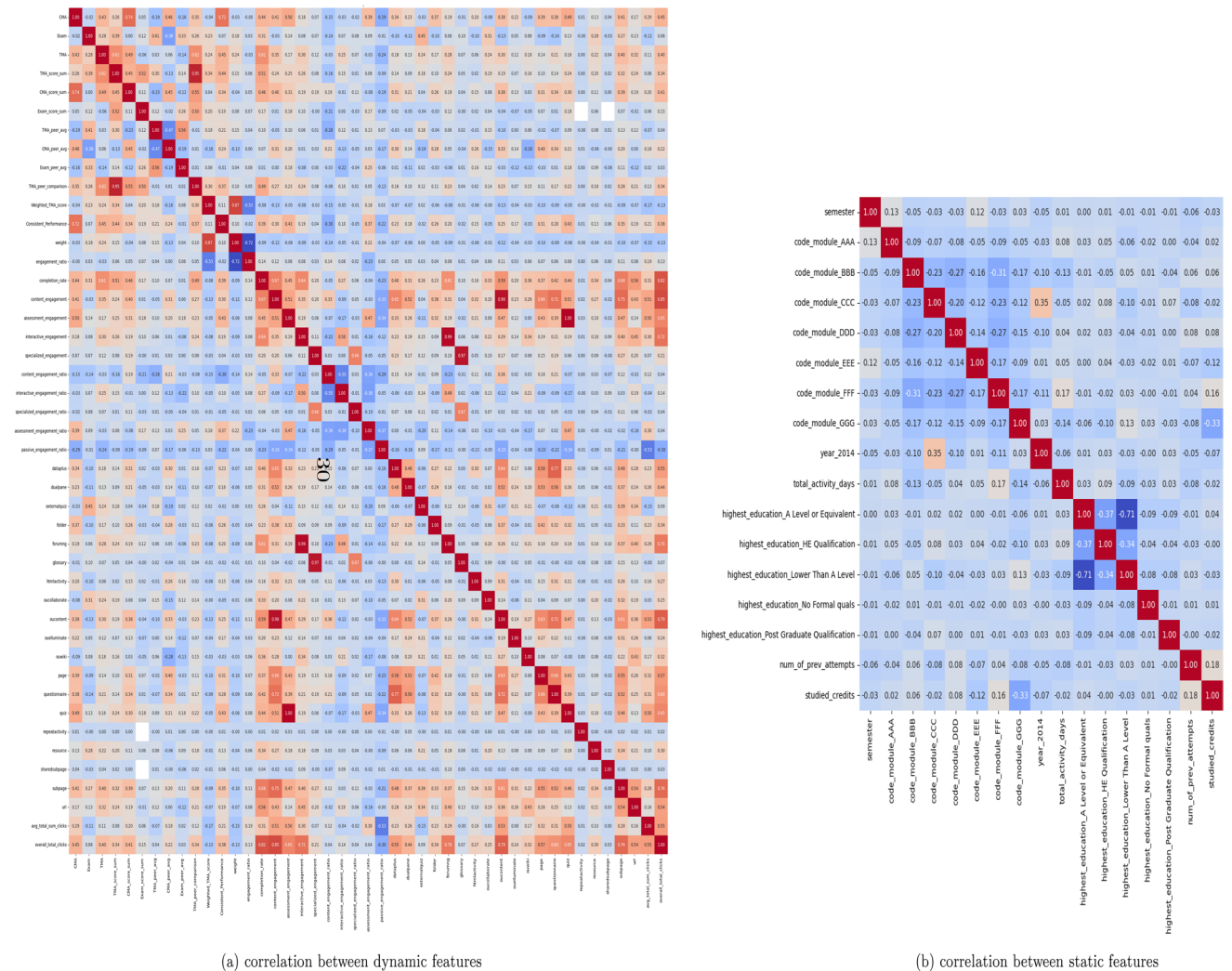
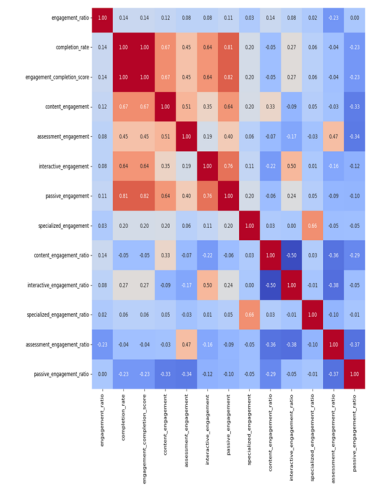
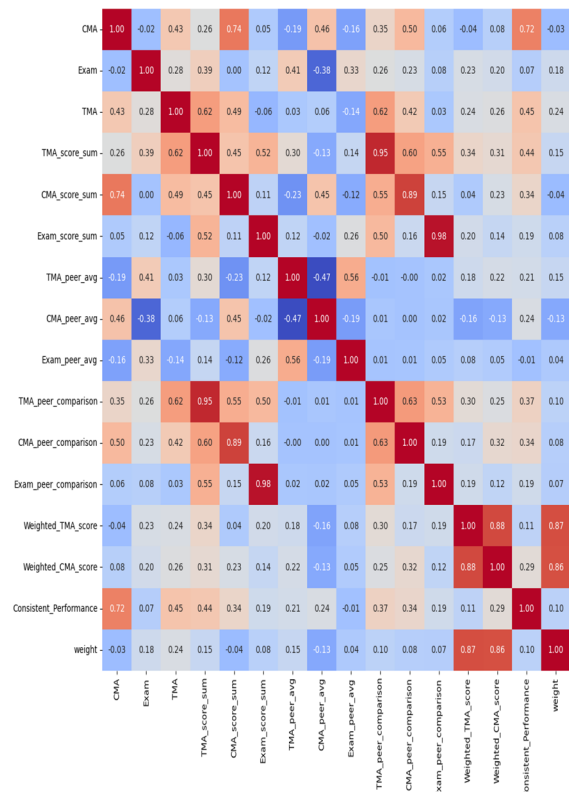


Fig. 7 Between groups correlation



(c) Engagement interaction correlation matrix



(e) Academic performance correlation matrix

Fig. 8 Within group correlation

Author contributions Conceptualization, B.A., M.B., M.A. and K.A.; methodology, B.A. and M.B.; validation, M.A. and K.A.; formal Analysis, B.A., M.B., M.A. and K.A.; resources, B.A., M.B. and M.A.; writing-original draft preparation, B.A.; writing-review & editing, M.B., M.A. and K.A.; proof reading B.A., M.B., M.A. and K.A.; supervision, M.B. and M.A.; funding acquisition, B.A. All authors have read and agreed to the published version of the manuscript.

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Consent to Participate Not applicable.

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